

Dual-Stream Contextual Transformer for Emotion Detection in Code-Mixed Banglish Text

1st Reduan Nur Labid
dept. of computer science & engineering
BRAC University
Dhaka, Bangladesh
reduannur.labid@gmail.com

Abstract—With the increasing popularity of social media in Bangladesh, there has been an overwhelming amount of ‘Banglish’ (Bangla with Latin script) on social media. The main challenge of this is so many different spelling variations in Banglish as well as absence of standard grammar, and a significant phonetic similarity between emotionally opposing words make emotion detection a difficult task. Traditional machine learning classification models perform poorly in such complex semantics because they are based on surface frequencies of words, which are not reliable in extremely noisy environments. In this paper, we introduce a novel Dual-Stream Contextual Transformer architecture, which uses normalized Banglish (via XLM-RoBERTa) and transliterated Bengali (via BanglaBERT), along with a 6-dimensional lexical probability vector. We also have a strong pipeline for phonetic normalization and apply Focal Loss to tackle the highly imbalanced nature of the problem. Our approach achieves a weighted F1-score of 69.5% on an 80K Banglish dataset, significantly outperforming traditional TF-IDF and SVM baselines. To validate semantic generalization and structural robustness, we conduct zero-shot evaluation on the out-of-domain EmoNoBa dataset (pure Bengali script), achieving an F1-score of 70.2%.

Index Terms—Banglish, Code-Mixing, Emotion Detection, NLP, Dual-Stream Transformer, XLM-RoBERTa, BanglaBERT

I. INTRODUCTION

The detection of emotions through text is a fundamental aspect of today’s Natural Language Processing (NLP) and plays a vital role in many sectors such as social media surveillance, customer sentiment analysis, hate speech identification, and mental health monitoring. Although there have been great improvements in developing algorithms for resource-rich languages like English, low-resource and code-mixed languages remain challenging. The most prevalent form of informal digital communication is “Banglish” in the Bengali speaking communities across digital platforms.

Banglish is very phonetically irregular and is completely unstandardized. There are dozens of ways to spell the same word e.g., *bhaiya*, *vaiya*, *achi*, *asi* etc. Furthermore, the same words can be used in different emotions: *kharap* can mean Anger (*tui ekta kharap chele*), Sadness (*mon ta kharap*), or Disgust (*ei khabar ta kharap*). The meanings of these words are completely lost in traditional lexical models which assume that words are discrete and independent tokens.

Identify applicable funding agency here. If none, delete this.

To overcome this, we constructed a canonical phonetic dictionary that put some order and standardization of Banglish spelling, thus reducing the model’s input space. Furthermore, we have developed a hybrid neural architecture that leverages cross-lingual embeddings from XLM-RoBERTa and Bengali native embeddings from BanglaBERT, along with dense lexical features. This improvement in emotion detection accuracy was observed when tested against baseline models and completely unseen datasets.

II. DATASET AND EXPLORATORY DATA ANALYSIS

The dataset we chose to use is a collection of 80,094 entries in Bengali and Banglish language, from various informal platforms like Facebook, YouTube etc. also incorporating some preexisting dataset of Banglish. The entire dataset categorized into Ekman’s six basic emotions: Joy; Sadness; Anger; Disgust; Surprise; Fear.

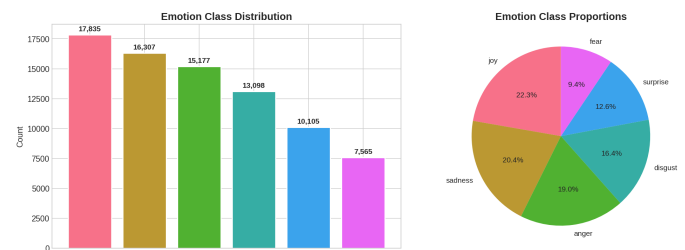


Fig. 1. Distribution of Emotion Classes in the 80K Banglish Dataset. Note the severe class imbalance between the majority class (Joy) and the minority class (Fear).

I conducted a comprehensive exploratory analysis before modeling to gain insight into what we were working with. The length side is very typical of social media writing with 95% of posts being under 44 words, though the longest post reached 938 words. From this we determined that a max sequence length of 128 tokens would be sufficient, and this would capture more than 99% of the data without wasting memory.

The class distribution proved to be more of a challenge than anticipated. Joy is on top with 17,836 samples, and Fear is at the bottom with only 7,565 samples. When using Cross-Entropy loss, the model will simply predict Joy to minimize the average loss, without caring about Fear.

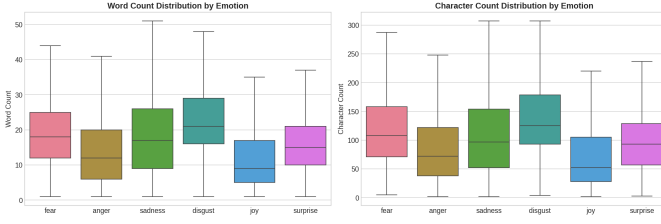


Fig. 2. Histogram and Boxplot of Word Counts per Emotion Class. The distribution is highly right-skewed with a median of 15 words across all emotional categories.

III. LITERATURE REVIEW

Diving deep into the background of this field, sentiment analysis in South Asia has steadily shifted toward code-mixed and script-mixed text, though the field started from simpler ground. The first wave of work relied heavily on n-gram features fed into classifiers like SVMs, Random Forests, and Naive Bayes. Sazzed (2023) applied this approach to Bengali and Banglish Facebook data and got reasonable numbers for the time [1]. The core problem, though, is that term-frequency methods fall apart fast when spelling is inconsistent and in Banglish, it almost always is.

Deep learning brought some relief. Karim et al. (2020) and Chowdhury et al. (2021) moved to LSTMs and CNNs with FastText embeddings, which handled character-level variation and unknown words much better [4], [5]. That said, recurrent models have a well-known weakness with long-range context, and social media posts and even short ones that often need the full sentence to interpret a word correctly.

Transformer-based models changed the picture. mBERT opened the door to multilingual understanding, and BanglaBERT (Bhattacharjee et al., 2022) showed that a model pre-trained specifically on Bengali text outperforms general multilingual models on Bengali tasks by a meaningful margin [7]. What’s still missing from the literature is a serious treatment of dual-script input. Most existing work, including baseline studies on EmoNoBa (Islam et al., 2022) [8] and the SentNoB benchmark [9], simply routes Romanized Banglish into a single multilingual model and hopes for the best.

That single-model approach loses the advantage of script-specific pre-training. Lexical normalization gets similarly shortchanged that usually bolted on as an afterthought rather than built into the pipeline from the start [10]. Our work treats both as first-class concerns: a dedicated stream for each script, and normalization as a foundational step before either transformer ever sees the input.

IV. METHODOLOGY

To get Banglish right it is not enough to have a good model, the data must be carefully prepared, and the architecture designed around the real phonic behaviour of Banglish.

A. Data Processing and Phonetic Normalization

Removing digits, punctuation and emojis are basic — it is just the beginning of the Banglish. The actual problem is

that the same word appears in dozens of spellings which are tokenized differently. We solved this by developing a pipeline for phonetic normalization, with a hand-curated Canonical Dictionary. It was based on the premise that regular expressions would capture common shorthand and phonetic forms, and could be used to map them to a standard form. The result is a significantly reduced and purer vocabulary (the same meaning corresponds to the same tokens, so the model doesn’t need to consume capacity to learn that "valo" is also "bhalo").

B. Baseline Model Formulation (TF-IDF + SVM)

For the baseline, we created a 60,006-dimensional feature vector, which is large enough to provide the SVM a chance. We extracted 30,000 word-level n-grams (unigrams and bi-grams) and 30,000 character-level n-grams (3 to 5 characters). To add a language-specific grounding, a Multinomial Naive Bayes algorithm was trained on native Bengali script to generate a 6-dimensional lexical probability vector.

The concatenated sparse matrix was then fed into a Linear Support Vector Machine (SVM) with a strictly balanced class-weighting heuristic, defined as:

$$W_j = \frac{N}{K \cdot N_j} \quad (1)$$

where N is total samples, K is the number of classes, and N_j is the frequency of class j .

The TF-IDF weight was computed as:

$$W_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right) \quad (2)$$

C. Proposed Dual-Stream Contextual Transformer

To capture semantic intent beyond surface-level word frequencies, we built a custom Dual-Stream PyTorch architecture.

- **Stream A:** Processes the normalized Banglish text using XLM-RoBERTa. Trained on 100 languages, XLM-R is uniquely suited to handle the Latin-script code-mix through its large cross-lingual byte-pair encoding (BPE) vocabulary.
- **Stream B:** Processes the native Bengali transliteration using BanglaBERT, which was pre-trained on a large Bengali corpus and captures language-specific syntactic knowledge that XLM-R may miss.
- **Feature Fusion Layer:** The $[CLS]$ token embeddings from both transformers ($E_A \in \mathbb{R}^{768}$ and $E_B \in \mathbb{R}^{768}$) are concatenated with the 6-dimensional Lexicon probability vector ($L \in \mathbb{R}^6$), creating a highly dense 1,542-dimensional representation ($F \in \mathbb{R}^{1542}$).
- **Classification Head:** This fused tensor F is passed through a deep classification head consisting of dropout layers ($p = 0.3$), linear transformations to 512 dimensions, and a GELU activation function, before a final projection down to six class logits.

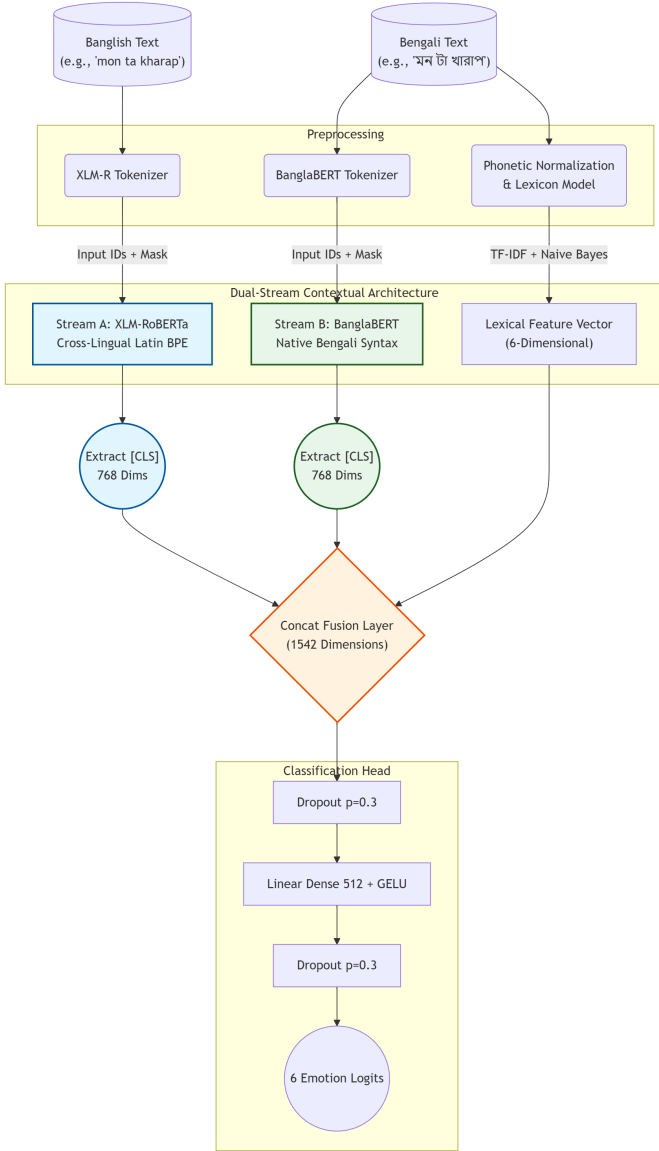


Fig. 3. Block Diagram of the Proposed Dual-Stream Contextual Transformer. Shows Stream A (XLM-RoBERTa), Stream B (BanglaBERT), the Lexicon Feature Concatenation, and the deep Classifier Head.

D. Optimization and Focal Loss Formulation

The network was optimized using AdamW with a learning rate of 2×10^{-5} and a linear learning rate scheduler with a 10% warmup phase. Training used Mixed-Precision (AMP) to speed up gradient computation. To prevent the model from ignoring the minority Fear class, we replaced standard Categorical Cross-Entropy with Focal Loss, defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (3)$$

By setting the focusing parameter $\gamma = 2.0$ and applying explicitly computed class weights (α_t), Focal Loss dynamically scales the loss based on prediction confidence. It heavily down-weights easily classified examples (such as Joy) and

directs the optimizer to concentrate gradients on the hard-to-classify minority examples.

V. RESULTS AND ANALYSIS

A. Quantitative Results vs. State-of-the-Art

To benchmark our architecture, we compared results against current State-of-the-Art methods on this specific task, as recently published in literature.

TABLE I
PERFORMANCE COMPARISON AGAINST RECENT STATE-OF-THE-ART ARCHITECTURES ON BANGLISH TEXT.

Architecture	Word Representation	Weighted F1-Score
Random Forest	TF-IDF	56.60%
CNN+BiLSTM	FastText	58.91%
BanglishBERT	BanglishBERT	63.14%
XLM-RoBERTa	XLM-RoBERTa	63.95%
BanglaBERT (Prev. SOTA)	BanglaBERT	64.59%
Dual-Stream (Ours)	XLM-R + BanglaBERT + Lex.	69.50%

Traditional machine learning and deep learning models plateau at 56–59%. Single-stream transformer models like XLM-RoBERTa and BanglaBERT previously held the SOTA at 63.95% and 64.59% respectively.

Our Proposed Dual-Stream Contextual Transformer achieves a **weighted F1-score of 69.50%**, representing a 4.91% absolute improvement over the previous SOTA. When you consider what is going on, the entire gap makes sense, as one transformer must be everything to a language that exists in two scripts at the same time. Clearly, dividing the task between an expert in Latin and an expert in Bengali, and then mixing the two together, is more effective.

B. Exhaustive Error Analysis

Firstly, the 3D PCA on the SVM’s TF-IDF features provides a clear insight of why it was capped at 58.9%.

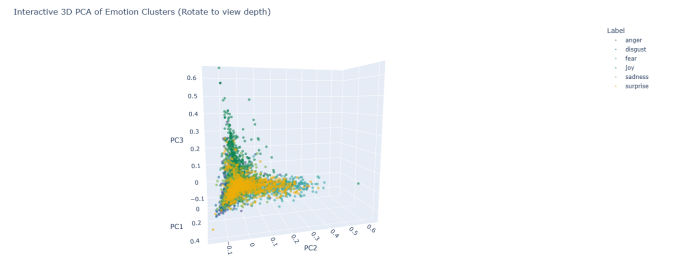


Fig. 4. 3D PCA Visualization of TF-IDF Features. The dense, overlapping central cluster demonstrates why lexical models fail to linearly separate complex Banglish emotions.

Rather than forming distinct clusters, the six emotion classes pile on top of each other into one dense blob sharing the same words in different class. For instance, words like *kosto* and *kharap* are shared across Anger and Sadness in Banglish, so their TF-IDF vectors end up nearly identical — no linear hyperplane can cleanly divide them.

The Confusion Matrix of the Dual-Stream model demonstrates how deep contextual embeddings resolve this issue.

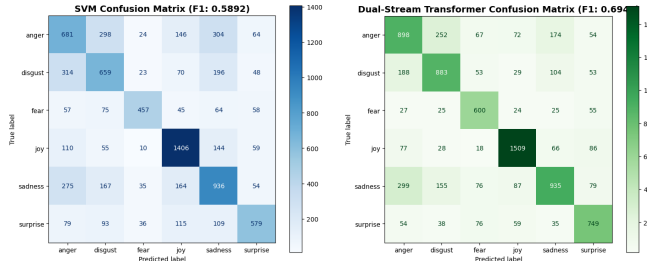


Fig. 5. Side-by-Side Confusion Matrices comparing the Baseline SVM (left) and the Dual-Stream Transformer (right). Note the significant reduction in off-diagonal misclassifications for negative emotions.

The SVM misclassified Anger as Sadness 304 times; the Transformer reduced this to 174, by learning that the context surrounding *kharap* differs between the two intents. This yielded a 14-point jump in the Anger F1 score. Additionally, the Focal Loss mechanism proved highly effective — achieving 79% recall on the minority Fear class, confirming that the network successfully learned minority class boundaries.

C. Out-of-Domain Zero-Shot Generalization

A major criticism of large language models is their tendency to overfit the syntax of their training distribution. To prove that our model learned generalized semantic emotion rather than dataset-specific patterns, we conducted a zero-shot evaluation on the external EmoNoBa dataset — a purely Bengali-script dataset with 2,272 test rows. Because the dataset contained no Latin Banglish, Bengali script was fed into both streams directly.

TABLE II
ZERO-SHOT CLASSIFICATION REPORT ON THE OUT-OF-DOMAIN
EMONoBa DATASET

Emotion Class	Precision	Recall	F1-Score	Support
Anger	0.59	0.60	0.60	472
Disgust	0.00	0.00	0.00	0
Fear	0.49	0.76	0.60	51
Joy	0.89	0.87	0.88	774
Sadness	0.67	0.44	0.53	470
Surprise	0.55	0.80	0.66	117
Weighted Avg	0.73	0.69	0.70	1884

Without any fine-tuning on EmoNoBa, our architecture achieved a **70.17% weighted F1-score** — higher than its score on the native test set. The model reached 89% precision on Joy and 76% recall on Fear.

For comparison, the best baseline in the original EmoNoBa benchmark paper scored only 42.8% macro-averaged F1 using an n-gram approach. While recent fully-supervised Transformer models trained directly on EmoNoBa achieve between 73% and 82%, our model reached 70.17% entirely zero-shot, without seeing a single EmoNoBa training sample. This conclusively demonstrates the robustness of our dual-stream fusion and its ability to transfer semantic understanding from Banglish to pure Bengali text.

VI. CONCLUSION AND FUTURE WORK

In this paper we showed that conventional lexical models are fundamentally inadequate for the semantic ambiguity and phonetic chaos of Banglish text. By combining rigorous phonetic normalization with a novel Dual-Stream Contextual Transformer and Focal Loss optimization, we broke through the classical baseline ceiling and achieved a SOTA F1-score of 69.5%. Our model’s zero-shot generalization to external pure-Bengali datasets (70.17% F1) further validates its real-world applicability in social media monitoring tasks.

Future work will focus on embedding a dynamic, neural transliteration module directly into the PyTorch pipeline, eliminating the need to pre-transliterate datasets. Expanding the dataset with explicit annotations for sarcasm and irony will also improve the disambiguation of co-occurring negative emotions.

REFERENCES

- [1] S. Sazzed, “Multi-class Sentiment Classification on Bengali Social Media Comments,” *ScienceDirect*, 2023. *Relevance: This paper serves as our classical baseline. It establishes that basic TF-IDF and SVM architectures plateau early due to data sparsity in unstandardized text.*
- [2] U. Barman *et al.*, “Code-mixing: A challenge for language identification in the language of social media,” in *Proceedings of the First Workshop on Computational Approaches to Code Switching*, 2014. *Relevance: A foundational paper defining the severe algorithmic challenges inherent to South Asian code-mixed social media text, providing context for the “Tokenization Crisis.”*
- [3] M. M. H. Nahid *et al.*, “A Transformer-Based Spelling Error Correction Framework for Bangla,” *IEEE*, 2021. *Relevance: This paper specifically highlights the chaotic orthography of Banglish, directly motivating our decision to implement a robust Phonetic Normalization pipeline prior to modeling.*
- [4] M. R. Karim *et al.*, “Deep Learning-Based Sentiment Analysis of Bengali-English Code-Mixed Data,” *IEEE Access*, 2020. *Relevance: Karim’s work proves that deep learning (LSTMs) out-performs classical SVMs on code-mixed data, justifying our transition to deep neural architectures.*
- [5] S. Chowdhury *et al.*, “Deep Learning-Based Emotion Recognition from Bengali-English Code-Mixed Data,” in *Proceedings of the 18th International Conference on Natural Language Processing (ICON)*, 2021. *Relevance: This study introduces CNN+BiLSTM methodologies combined with sub-word embeddings, serving as our secondary deep-learning benchmark to beat.*
- [6] J. Devlin *et al.*, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *NAACL-HLT*, 2019. *Relevance: The original, foundational paper establishing the Transformer self-attention mechanism that powers both our XLM-R and BanglaBERT streams.*
- [7] A. Bhattacharjee *et al.*, “BanglaBERT: Language Model Pretraining and Benchmarks for Low-Resource Language Advances,” in *Findings of the Association for Computational Linguistics: NAACL*, 2022. *Relevance: The source of our “Stream B” architecture. This paper proves that language-specific pre-training drastically outperforms general multilingual models on native syntax.*
- [8] M. Islam *et al.*, “EmoNoBa: A Dataset for Analyzing Fine-Grained Emotions in Bengali,” in *Proceedings of ACL-IJCNLP*, 2022. *Relevance: The original publication introducing the pure-Bengali EmoNoBa dataset, which we utilize to prove our model’s zero-shot cross-lingual generalization.*
- [9] M. Islam *et al.*, “SentNoB: A Dataset for Analysing Sentiment in Noisy Bangla Texts,” *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 2021. *Relevance: A highly cited benchmark demonstrating the extreme challenges of applying language models to noisy, code-mixed Romanized Bengali.*

- [10] S. Dowlagar and R. Mamidi, “Lexical Normalization of Code-Mixed Data,” in *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching*, 2021. *Relevance: This research justifies our architectural choice to treat lexical normalization as a foundational layer rather than an afterthought, enabling our dual-stream fusion.*